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In this problem, original inputs are linearly inseparable
i.e. there is no straight line possible which can separate
the two classes.

So, ^{one} hidden neuron draws one straight line
 $w_1x_1 + w_2x_2 + b = 0 \rightarrow$ straight line in x_1-x_2 plane

Now we put (x_1, x_2) in it and check the point
lies in which side. If $z = w_1x_1 + w_2x_2 + b > 0$
then neuron becomes active else inactive

So the hidden neuron transforms the problem into
new representation where classes are not mixed.

So using hidden layers we ^{can} classify by reshaping the input
space into one where problem becomes linearly separable